USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] **[23ESCE12]**

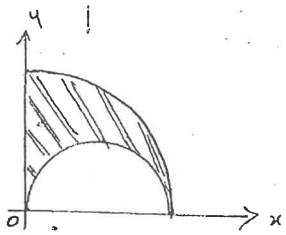
FIRST SEMESTER B.Tech EXAMINATION, JANUARY- 2024

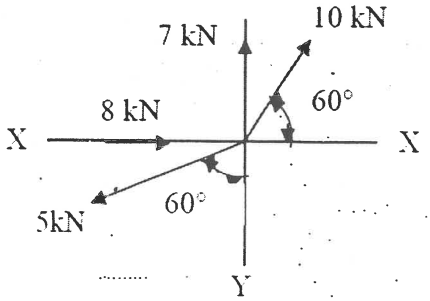
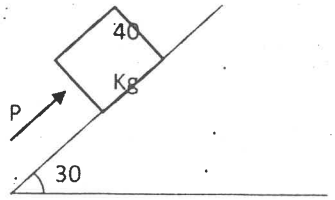
ENGINEERING MECHANICS

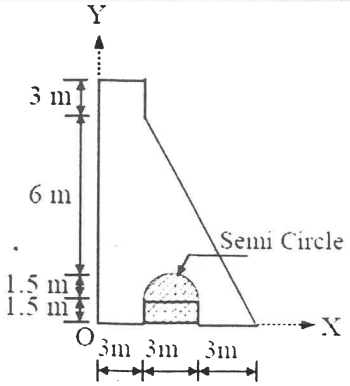
Time: 3 Hours **Maximum Marks: 100**

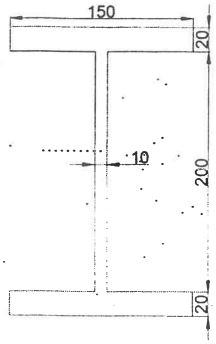
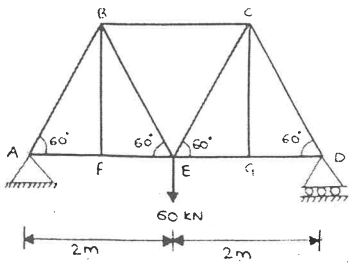
Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	**a.**	What is coplanar concurrent forces.	L-2	CO1	**02**
	b.	Define co-efficient of friction.	L-1	CO2	**02**
	c.	Define center of gravity.	L-1	CO3	**02**
	d.	State perpendicular axis theorem.	L-1	CO4	**02**
	e.	Write the equations of static equilibrium.	L-2	CO5	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	State and prove parallelogram law of forces theorem.	L-3	CO1	**05**
		OR			
	b.	Explain the following with neat sketch **i.** Principle of superposition of forces **ii.** Principle of transmissibility of forces	L-3	CO1	**05**
3.	**a.**	State laws of static friction.	L-3	CO2	**05**
		OR			
	b.	What is plane frame and space frame?	L-3	CO2	**05**
4.	**a.**	Prove that centroid of rectangle is at the centre.	L-4	CO3	**05**
		OR			
	b.	Find the centroid of the shaded area by cutting a semicircle of diameter 100mm from a quadrant of circle of radius 100mm, as shown in the figure.  	L-4	CO3	**05**
5.	**a.**	State and prove parallel axis theorem	L-4	CO4	**05**
		OR			
	b.	Find the moment of inertia of a rectangular section.	L-4	CO4	**05**
6.	**a.**	A particle moves along a straight line such that its position is defined by $s = (t^3 - 3t^2 + 2)$ m. Determine the velocity and acceleration of the particle when t = 4s	L-4	CO5	**05**
		OR			

	b.	Explain displacement-time graphs.	L-4	CO5	**05**
7.	**a.**	With the help of sketches explain five types of force system.	L-3	CO6	**05**
		OR			
	b.	Solve the relation involving displacement, velocity and acceleration with time as constant.	L-4	CO6	**05**
		PART-C	**06 X 10=60**		
8.	**a.**	List the types of beams and explain each with a neat sketch.	L-3	CO1	**10**
		OR			
	b.	Determine the magnitude and direction of the resultant along with the components of the forces made by the given force system. 	L-4	CO1	**10**
9.	**a.**	A 40 Kg mass is placed on the inclined plane making angle of 30 with horizontal as shown in the figure. A push "P" is applied parallel to the plane. If co-efficient of static friction between the plane & the mass is 0.25, find the maximum & minimum value of P between which the mass will be in equilibrium. 	L-4	CO2	**10**
		OR			
	b.	Write short note on the following **i.** Limiting friction **ii.** Angle of friction	L-3	CO2	**10**
10.	**a.**	Find the centroid of L-section with web dimensions 40mmX200mm and flange dimensions 200mmX50mm.	L-4	CO3	**10**
		OR			
	b.	Determine the co-ordinates of centroid with respect to 'O' of the section as shown in the figure.	L-4	CO3	**10**



11.	**a.**	Determine the moment of inertia with respect to both X & Y axes for I-section with top flange dimensions of 24mm X 6mm, bottom flange dimensions of 48mm X 6mm and web dimension of 8mm X 48mm.	L-4	CO4	**10**
		OR			
	b.	Find the moment of inertia for the symmetrical I-section shown in the figure about axis passing through its centroid 	L-4	CO4	**10**
12.	**a.**	A car moves in a straight line such that for a short time its velocity is defined by v = (5t^2 + 4t) m/s, where t is in seconds. Determine its position and acceleration when t = 4 s. (When t = 0, s = o).	L-4	CO5	**10**
		OR			
	b.	Derive the relation between displacement, velocity and acceleration with uniform acceleration.	L-4	CO5	**10**
13.	**a.**	Find the forces acting in all members of the truss shown in figure 	L-4	CO2	**10**
		OR			
	b.	Show from first principles the centroid of a triangle lies at h/3 from the base b.	L-5	CO6	**10**